MAT205: Abstract Algebra II

2. Group Actions

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What is a Group Action?

Definition. A (left) **group action** of G on a set X is a map $G \times X \to X$, written $(g, x) \mapsto g \cdot x$, satisfying:

1. **Identity**: $e \cdot x = x$ for all $x \in X$
2. **Compatibility**: $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$, $x \in X$

Equivalently, a group action is a homomorphism $\varphi : G \to \mathrm{Sym}(X)$.

Each $g \in G$ defines a **bijection** $x \mapsto g \cdot x$ on X.

Key vocabulary:

- **Transitive**: $\forall x, y \in X,\ \exists g \in G$ such that $g \cdot x = y$ (only one orbit)
- **Faithful**: only e fixes every x (φ is injective)
- **Free**: $g \cdot x = x$ for some $x \Rightarrow g = e$ (no non-trivial fixed points)

Group Actions $\leftrightarrow$ Homomorphisms

A group action $G \curvearrowright X$ is **the same thing** as a homomorphism $\varphi : G \to \mathrm{Sym}(X)$.

Action $\to$ homomorphism: Given $G \curvearrowright X$, define $\varphi(g)(x) = g \cdot x$. Then $\varphi(g)$ is a bijection $X \to X$, and $\varphi(gh) = \varphi(g) \circ \varphi(h)$.

Homomorphism $\to$ action: Given $\varphi : G \to \mathrm{Sym}(X)$, define $g \cdot x = \varphi(g)(x)$.

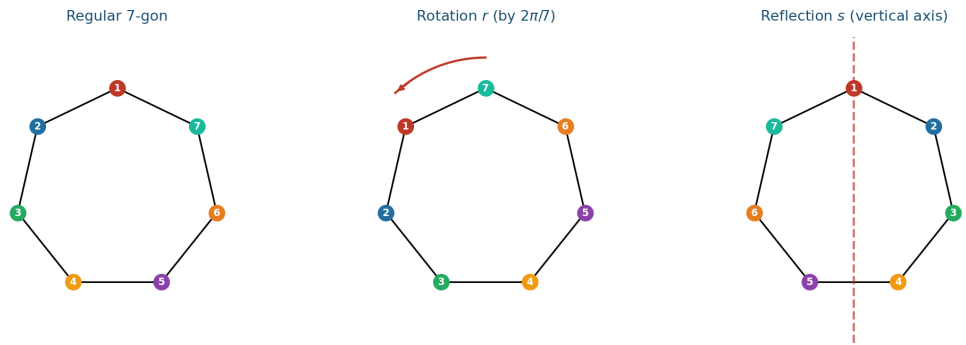
Under φ: **transitive** $\leftrightarrow$ single orbit, **faithful** $\leftrightarrow$ φ injective, **free** $\leftrightarrow$ no non-trivial fixed points.

Cayley's Theorem. Every group G embeds into $\mathrm{Sym}(G)$ via the left-multiplication action. So every group is isomorphic to a subgroup of some symmetric group.

Cayley (1821–1895)

Discrete Examples I: Symmetry of Polygons

Example 1. The dihedral group D_n acts on the vertices of a regular n-gon.



D_7 has $2 \times 7 = 14$ elements: 7 rotations $(e, r, r^2, \ldots, r^6)$ and 7 reflections.

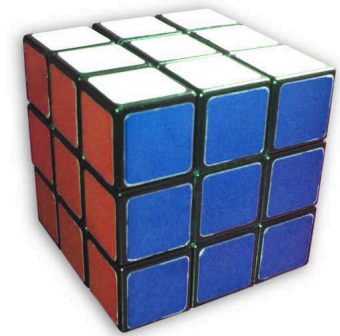
Each symmetry permutes the 7 vertices $\implies D_7 \hookrightarrow S_7$.

Example 2. The **Rubik's cube group** acts on the facelets of a Rubik's cube.

The Rubik's Cube Group

In the language of group actions:

Group G	$\langle U, D, L, R, F, B \rangle$ (generated by 6 face rotations)
Set X	48 movable facelets (exclude 6 centers)
Action	$G \leq S_{48}$, each move permutes facelets



Faithful? Yes – only e fixes all 48 facelets.

Transitive? No – two orbits: 24 corner facelets (8 cubies $\times$ 3) and 24 edge facelets (12 cubies $\times$ 2).

Order: $|G| = \frac{8! \cdot 3^8 \cdot 12! \cdot 2^{12}}{12} \approx 4.3 \times 10^{19}$ (3 constraints: parity, twist, flip).

God's number = 20: diameter of Cayley graph (Rokicki et al., 2010).

Discrete Examples II: Conjugation

Example 3. Any group G acts on **itself** by conjugation:

$$g \cdot x = gxg^{-1}$$

- **Orbit** of x = **conjugacy class** $\mathrm{Cl}(x) = \{gxg^{-1} \mid g \in G\}$
- **Stabilizer** of x = **centralizer** $C_G(x) = \{g \in G \mid gx = xg\}$
- **Fixed points** = **center** $Z(G) = \{x \in G \mid gx = xg \text{ for all } g\}$

This action is the key tool for Sylow theorems (next lectures).

Conjugacy Classes of $GL_n(\mathbb{C})$

$A, B \in GL_n(\mathbb{C})$ are conjugate $\iff$ they have the **same Jordan normal form** (up to block order).

A **Jordan block**: $J_m(\lambda) = \begin{pmatrix} \lambda & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{pmatrix} \in GL_m(\mathbb{C})$, where $\lambda \in \mathbb{C}^\times$.

Every A is conjugate to $J_{n_1}(\lambda_1) \oplus \cdots \oplus J_{n_k}(\lambda_k)$ (unique up to reordering).

Conjugacy classes $\longleftrightarrow$ **Jordan types** (partitions of n labeled by nonzero eigenvalues):

n	Jordan type	Description
1	(λ)	one class per $\lambda \in \mathbb{C}^\times$
2	$\mathrm{diag}(\lambda, \mu)$, $\lambda \neq \mu$	diagonalizable, distinct eigenvalues
2	λI_2	scalar matrix – **central** in GL_2
2	$J_2(\lambda)$	non-diagonalizable

Discrete Examples III: Action on Cosets

Example 4. G acts on the coset space G/H by left multiplication:

$$g \cdot (aH) = (ga)H$$

This is always a **transitive** action (only one orbit).

Stabilizer of the coset eH: $G_{eH} = H$.

Example 5. S_n acts on polynomials by permuting variables:

$$\sigma \cdot f(x_1, \ldots, x_n) = f(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$$

The **fixed points** are the **symmetric polynomials** (e.g., $e_1 = \sum x_i$, $e_2 = \sum_{i<j} x_i x_j$).

Orbits

Definition. The **orbit** of $x \in X$ under G is:

$$G \cdot x = \{g \cdot x \mid g \in G\}$$

Theorem. The orbits partition X:

$$X = \bigsqcup_{i \in I} G \cdot x_i$$

where $\{x_i\}_{i \in I}$ is a set of orbit representatives (the index set I may be infinite).

Proof. Define $x \sim y \iff \exists\, g \in G,\ g \cdot x = y$. This is an equivalence relation:

- Reflexive: $e \cdot x = x$
- Symmetric: $g \cdot x = y \implies g^{-1} \cdot y = x$
- Transitive: $g \cdot x = y,\ h \cdot y = z \implies (hg) \cdot x = z$ $\square$

Stabilizers

Definition. The **stabilizer** (or **isotropy group**) of $x \in X$ is:

$$G_x = \{g \in G \mid g \cdot x = x\}$$

Proposition. $G_x \leq G$ (it is a subgroup).

Proof. (i) $e \cdot x = x$, so $e \in G_x$. (ii) If $g, h \in G_x$, then $(gh) \cdot x = g \cdot (h \cdot x) = g \cdot x = x$. (iii) If $g \in G_x$, then $g^{-1} \cdot x = g^{-1} \cdot (g \cdot x) = (g^{-1}g) \cdot x = x$. $\square$

Action	Stabilizer of x	Name
Conjugation on G	$C_G(x)$	centralizer
Left mult. on G/H	H	the subgroup itself
$SO(3)$ on S^2	rotations fixing $x \cong SO(2)$	–

The Orbit-Stabilizer Theorem

Theorem. If G acts on X and $x \in X$, then

$$|G \cdot x| = [G : G_x] = \frac{|G|}{|G_x|}$$

Proof sketch. The map $gG_x \mapsto g \cdot x$ is a well-defined bijection from G/G_x to $G \cdot x$.

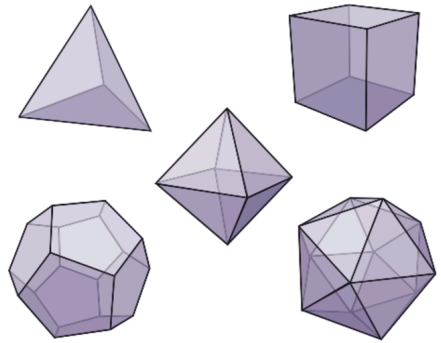
Example: D_4 acts on vertices of a square.

- Orbit of vertex 1 = all 4 vertices (transitive), so $|G \cdot 1| = 4$
- $|D_4| = 8$, so $|G_1| = 8/4 = 2$
- Indeed, $G_1 = \{e, s\}$ where s is the reflection fixing vertex 1

Corollary. If G is finite and acts on X, then $|G \cdot x|$ divides $|G|$.

The Platonic Solids

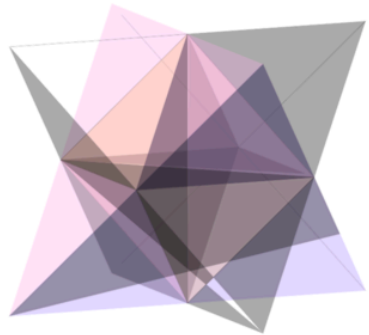
The five **Platonic solids** – the only convex regular polyhedra in $\mathbb{R}^3$.



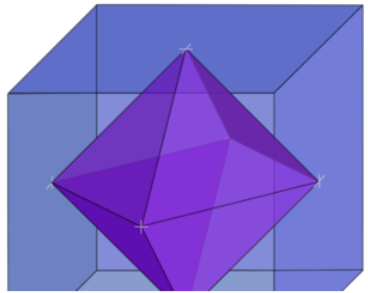
Dual pair	Rotation group	Full symmetry
Tetrahedron (self-dual)	A_4 (12)	S_4 (24)
Cube $\leftrightarrow$ Octahedron	S_4 (24)	$S_4 \times C_2$ (48)
Dodecahedron $\leftrightarrow$ Icosahedron	A_5 (60)	$A_5 \times C_2$ (120)

Duality of Platonic Solids

The **dual** is formed by placing a vertex at each **face center**. Dual pairs share the same symmetry group.



Tetrahedron (self-dual)



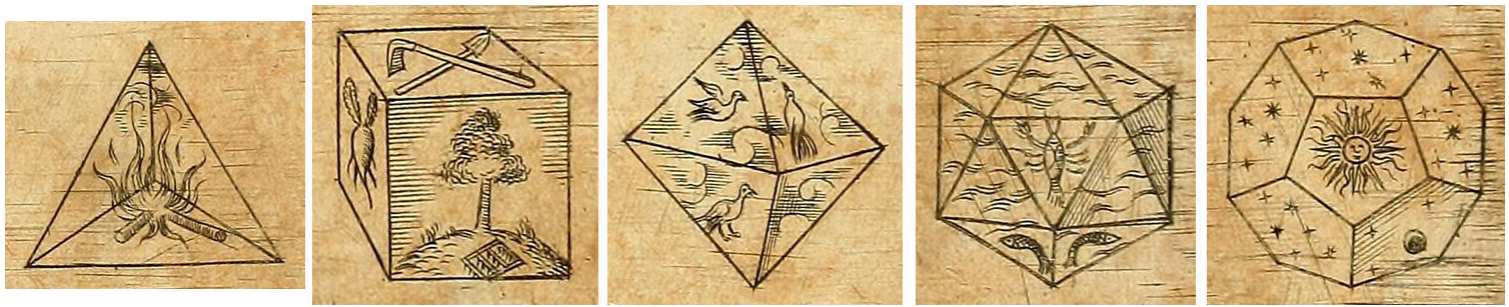
Cube <-> Octahedron

- Tetrahedron $\leftrightarrow$ Tetrahedron (self-dual: $V = F = 4$)
- Cube ($V = 8, F = 6$) $\leftrightarrow$ Octahedron ($V = 6, F = 8$): vertices $\leftrightarrow$ faces
- Dodecahedron ($V = 20, F = 12$) $\leftrightarrow$ Icosahedron ($V = 12, F = 20$)

$A_5 \cong$ rotation group of the icosahedron – the smallest **non-abelian simple group**. This connects to the **insolvability of the quintic** (Part II).

Kepler's Harmonice Mundi (1619)

Kepler assigned each Platonic solid to a **classical element**:



Tetrahedron	Cube	Octahedron	Icosahedron	Dodecahedron
Fire	Earth	Air	Water	Cosmos
sharpest	most stable	"freely suspended"	most fluid	12 faces = zodiac

Kepler also used the Platonic solids to model **planetary orbits** in his *Mysterium Cosmographicum* (1596) – nesting each solid between planetary spheres. Though wrong as astronomy, it drove him toward his three laws of planetary motion.

The Orthogonal and Special Orthogonal Groups

Definition. The **orthogonal group** $O(n) = \{A \in GL_n(\mathbb{R}) \mid A^T A = I\}$ – the group of distance-preserving linear maps.

The **special orthogonal group** $SO(n) = \{A \in O(n) \mid \det(A) = 1\}$ – the **rotation** group.

- $SO(2) = \left\{ \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \middle| \, \theta \in [0, 2\pi) \right\} \cong S^1$ – rotations of the plane
- $SO(3)$ – rotations of $\mathbb{R}^3$. Every element is a rotation about some axis (Euler's rotation theorem).
- $O(n) = SO(n) \sqcup \{\text{reflections}\}$: $[O(n) : SO(n)] = 2$

Continuous Examples I: $SO(3)$ on S^2

Example 6. $SO(3)$ acts on S^2 – **transitively** (one orbit = entire sphere).

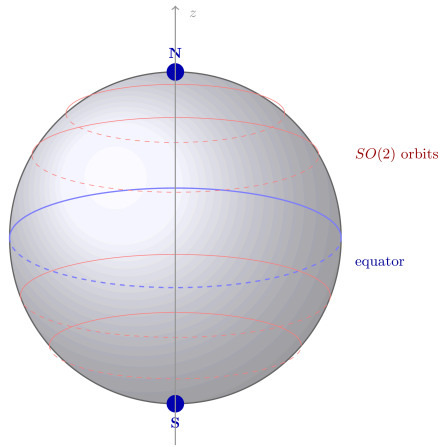
- **Stabilizer** of north pole N: rotations about z-axis $= SO(2)$
- **Orbit-stabilizer**: $S^2 \cong SO(3)/SO(2)$

The subgroup $SO(2) \leq SO(3)$ acts on S^2 **non-transitively**:

- **Orbits** = latitude circles (red in figure)
- **Fixed points** = north and south poles

Example 7. $O(n) \curvearrowright \mathbb{R}^n$ (orthogonal transformations).

- **Orbits** = concentric spheres S_r^{n-1}
- **Stabilizer** of $(r, 0, ..., 0) \cong O(n-1)$



$SO(3)$: transitive (one orbit = S^2)

$SO(2)$: orbits = latitude circles

Continuous Examples II: Linear and Modular

Example 8. $GL_n(\mathbb{R})$ acts on $\mathbb{R}^n$ by matrix multiplication: $A \cdot v = Av$.

- **Orbits**: $\{0\}$ and $\mathbb{R}^n \setminus \{0\}$ (any nonzero vector maps to any other)
- **Stabilizer** of $e_1 = (1, 0, ..., 0)^T$: matrices with first column e_1

Example 9. $SL_2(\mathbb{Z})$ acts on the upper half-plane $\mathbb{H}$ (from Lecture 1):

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}$$

- **Orbit space** $\mathbb{H}/SL_2(\mathbb{Z})$ = modular curve (parameterizes elliptic curves)
- **Stabilizer** of $z = i$: $\{\pm I, \pm\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)\} \cong \mathbb{Z}/4\mathbb{Z}$ (in $PSL_2(\mathbb{Z})$: $\mathbb{Z}/2\mathbb{Z}$)

Klein's Erlangen Program (1872)

Felix Klein proposed: **a geometry is a group action**.

A **geometry** = a set X together with a group G acting on X. The "geometric properties" of X are exactly the properties invariant under G.

Geometry	Space X	Group G	Invariants
Euclidean	$\mathbb{R}^n$	$O(n) \ltimes \mathbb{R}^n$	distances, angles
Affine	$\mathbb{R}^n$	$GL_n \ltimes \mathbb{R}^n$	parallelism, ratios
Projective	$\mathbb{P}^n$	PGL_{n+1}	cross-ratio, collinearity
Hyperbolic	$\mathbb{H}^n$	$SO(n,1)$	hyperbolic distance

"More symmetry = fewer invariants = coarser geometry."

This viewpoint unifies all of classical geometry under group theory – and motivates everything in this course.

The Class Equation

Apply conjugation action $G \curvearrowright G$ with $g \cdot x = gxg^{-1}$:

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)]$$

where the sum runs over representatives of non-central conjugacy classes.

Example: S_3 has order 6, center $Z(S_3) = \{e\}$.

Conjugacy class	Size	$[G : C_G(x)]$
$\{e\}$	1	(center)
$\{(12), (13), (23)\}$	3	$6/2 = 3$
$\{(123), (132)\}$	2	$6/3 = 2$

Application: Center of a p-Group

Theorem. If $|G| = p^n$ with p prime and $n \geq 1$, then $Z(G) \neq \{e\}$.

Proof. Apply the **class equation**:

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)]$$

Each $[G : C_G(x_i)]$ divides $|G| = p^n$ and is > 1, so $p \mid [G : C_G(x_i)]$.

Therefore $p \mid |G| - \sum [G : C_G(x_i)] = |Z(G)|$.

Since $e \in Z(G)$, we have $|Z(G)| \geq p > 1$. $\square$

Corollary. Groups of order p^2 are abelian.

Homework. Prove: if $G/Z(G)$ is cyclic, then G is abelian. Use this to prove the corollary.

Burnside's Lemma

Theorem (Burnside / Cauchy–Frobenius). The number of orbits of G acting on X is:

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

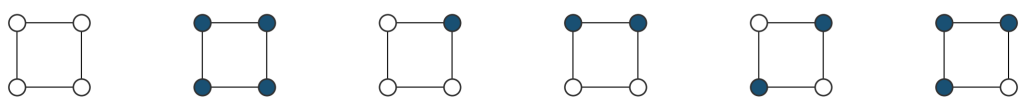
where $X^g = \{x \in X \mid g \cdot x = x\}$ is the set of fixed points of g.

Proof. Count the set $S = \{(g, x) \in G \times X \mid g \cdot x = x\}$ in two ways:

- By g: $|S| = \sum_{g \in G} |X^g|$
- By x: $|S| = \sum_{x \in X} |G_x| = \sum_{\text{orbits } O} \sum_{x \in O} \frac{|G|}{|O|} = |X/G| \cdot |G| \; \square$

Q: How many distinct ways to color the vertices of a square with 2 colors, up to rotation and reflection?

Distinct colorings of a square (2 colors, up to D_4)



Burnside: Square Coloring

Vertices $1, 2, 3, 4$ clockwise. $D_4 = \langle r, s \rangle$, $r = (1234)$, $s = (24)$. Color each vertex B or W. $|X^g| = 2^{(\text{number of cycles})}$.

g	Permutation	Cycles	$\lvert X^g \rvert$
e	id	$(1)(2)(3)(4)$ – 4 cycles	16
r, r^3	(1234), (1432)	1 cycle each	2, 2
r^2	$(13)(24)$	2 cycles	4
s	(24)	$(1)(3)(24)$ – 3 cycles	8
$r^2 s$	(13)	$(2)(4)(13)$ – 3 cycles	8
$rs, r^3 s$	$(12)(34)$, $(14)(23)$	2 cycles each	4, 4

$$|X/G| = \frac{1}{8}(16 + 2 + 4 + 2 + 8 + 4 + 8 + 4) = \frac{48}{8} = 6$$

Homework (Lecture 2)

Problem 1. Bracelets with 7 beads

Let X be the set of all colorings of the vertices of a regular heptagon using two colors. Let D_7 act on X by permuting the vertices.

a. Compute $|X|$.

b. How many colorings are fixed by the identity element?

c. How many colorings are fixed by a nontrivial rotation?

d. How many colorings are fixed by a reflection?

e. Use Burnside's lemma to compute the number of distinct bracelets.

Problem 2. The sphere as a homogeneous space

a. Prove that $SO(3)$ acts on the unit sphere $S^2 \subset \mathbb{R}^3$.

b. Prove that this action is transitive.

c. Let $N = (0, 0, 1)$ be the north pole. Compute the stabilizer $\mathrm{Stab}(N)$.

Looking Ahead

We now have the tools for **Sylow theory**:

- **Lecture 3**: Cauchy's theorem (via group action on tuples) and the normalizer lemma
- **Lecture 4**: Sylow I (existence), Sylow II (conjugacy), Sylow III (counting)

The key idea: choose the right group action on the right set, then count.

"Group actions are the single most important tool in finite group theory."

Questions?